ADR IDENTIFICATION BASICS

Educational Pharmacovigilance Case-Analysis Report

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Course: B.Pharm

Subject: Pharmacovigilance / ADR Identification

Task ID: task1

Important note: This report uses fictional teaching cases because the screenshot shows the task description but not the platform's actual patient scenarios.

1. Introduction

An adverse drug reaction (ADR) is a harmful and unintended response to a medicinal product when it is used at normally intended doses for prevention, diagnosis, or treatment, or for modification of a physiological function. Pharmacovigilance is the science and activities relating to the detection, assessment, understanding, and prevention of adverse effects or any other medicine-related problem.

2. Objectives

• Review symptoms and medication history.

• Establish the time relationship between medicine use and the event.

• Identify the suspected medicine and consider alternative explanations.

• Record missing information and suggest appropriate follow-up.

3. Method used for case analysis

1. Identify the event without assuming its cause.

2. Record suspected medicine, dose, route, indication, and dates if available.

3. Check whether the event began after the medicine was started.

4. Consider other explanations such as disease, infection, other medicines, or food.

5. Review dechallenge/rechallenge only if already available; never deliberately restart a suspected medicine to test causality.

6. Assess seriousness, outcome, and missing information; document uncertainty.

A short case description is not enough to prove that a medicine caused an event. Formal causality assessment requires complete clinical information and an appropriate method.

4. Case 1 – Suspected skin rash after amoxicillin

Patient scenario

A 25-year-old patient was prescribed amoxicillin for a suspected bacterial infection. Two days after starting treatment, the patient noticed a red, itchy skin rash. The patient's previous allergy history and other medicines are not provided.

Assessment item	Case finding
Suspected medicine	Amoxicillin
Reported reaction	Red, itchy skin rash
Time relationship	Reaction began about 2 days after starting the medicine.
Preliminary analysis	The timing is compatible with a possible drug reaction, but the information is insufficient to confirm causality. Infection-related rash, other medicines, and allergy history must be considered.
Preliminary category	Possible / unassessable with the limited information provided
Information still required	Dose, route, indication confirmation, start/stop dates, previous penicillin allergy, other medicines, clinical examination, outcome after stopping, and any alternative cause.
Suggested follow-up / safety note	Do not self-rechallenge. The patient should contact the prescriber promptly for assessment. Urgent medical help is needed for breathing difficulty, facial/throat swelling, faintness, blistering, or a rapidly worsening rash.

5. Case 2 – Gastrointestinal symptoms after metformin

Patient scenario

A 48-year-old patient with type 2 diabetes started metformin. During the first week, the patient experienced nausea and loose stools. No information is available about dose, formulation, food intake, severity, or other causes.

Assessment item	Case finding
Suspected medicine	Metformin
Reported reaction	Nausea and loose stools
Time relationship	Symptoms appeared during the first week after treatment began.
Preliminary analysis	Gastrointestinal symptoms are known to occur with metformin, and the timing may fit. However, infection, diet, dose, and other medicines should be reviewed before assigning causality.
Preliminary category	Possible (based on the limited case description)
Information still required	Dose and titration schedule, immediate- or extended-release formulation, relation to meals, hydration, other medicines, infection history, severity, and outcome.
Suggested follow-up / safety note	Discuss symptoms with the prescriber or pharmacist. Do not stop or change diabetes treatment without professional advice. Seek urgent care for severe dehydration, persistent vomiting, or serious deterioration.

6. Case 3 – Stomach irritation after ibuprofen

Patient scenario

A 35-year-old patient took ibuprofen for pain. Later that day, the patient reported upper-abdominal discomfort and acidity. The dose, previous ulcer history, alcohol use, and concurrent medicines are unknown.

Assessment item	Case finding
Suspected medicine	Ibuprofen
Reported reaction	Upper-abdominal discomfort / dyspepsia
Time relationship	Symptoms were reported later on the day ibuprofen was taken.
Preliminary analysis	A temporal relationship is present and gastrointestinal irritation is a recognised effect of NSAIDs. Other causes and risk factors have not been ruled out, so the relationship remains suspected rather than confirmed.
Preliminary category	Possible / insufficient information for a firm assessment
Information still required	Dose, frequency, food intake, history of ulcer or bleeding, anticoagulants/steroids/other NSAIDs, alcohol use, symptom severity, and outcome.
Suggested follow-up / safety note	The patient should seek advice from a healthcare professional before taking further doses. Black stools, vomiting blood, severe abdominal pain, or fainting require urgent medical attention.

7. Case 4 – Drowsiness after cetirizine

Patient scenario

A 22-year-old patient took cetirizine for allergic symptoms and felt sleepy several hours later. The patient had not reported this symptom before taking the dose. Other medicines and sleep status are not known.

Assessment item	Case finding
Suspected medicine	Cetirizine
Reported reaction	Drowsiness / sleepiness
Time relationship	Sleepiness occurred several hours after a dose.
Preliminary analysis	The timing and known sedating potential of cetirizine support a suspected association. Sleep deprivation, alcohol, and other sedating medicines may also contribute.
Preliminary category	Possible; further information would strengthen the assessment
Information still required	Dose, exact onset and duration, other sedatives, alcohol, sleep pattern, medical history, and whether symptoms resolved.
Suggested follow-up / safety note	Avoid driving or operating machinery while drowsy. Ask a pharmacist or prescriber for advice, especially if drowsiness is severe or persistent.

8. Case 5 – Low blood glucose after insulin

Patient scenario

A 60-year-old patient using insulin developed sweating, shakiness, and confusion after taking insulin but delaying a meal. A blood glucose value and insulin dose are not supplied.

Assessment item	Case finding
Suspected medicine	Insulin (specific product and dose not stated)
Reported reaction	Suspected hypoglycaemia; symptoms include sweating, shakiness, and confusion
Time relationship	Symptoms occurred after insulin administration and a delayed meal.
Preliminary analysis	The sequence is consistent with insulin-associated hypoglycaemia, especially with a delayed meal. A measured glucose level and full clinical details are needed to assess the event.
Preliminary category	Possible; clinically important suspected reaction
Information still required	Measured blood glucose, insulin type/dose/time, meal details, exercise, alcohol, renal function, other glucose-lowering medicines, treatment given, and outcome.
Suggested follow-up / safety note	If the patient is conscious and able to swallow, follow their clinician-provided hypoglycaemia plan. If unconscious, having a seizure, or unable to swallow, give nothing by mouth and seek emergency help. Review the insulin/meal plan with the diabetes team.

9. Overall conclusion

The five fictional cases demonstrate how a suspected ADR is identified by reviewing the medicine, reported event, and time relationship. Alternative causes and missing clinical details must also be considered. A temporal association alone does not prove causation. Suspected reactions should be documented and assessed by an appropriately qualified healthcare professional, and serious or urgent symptoms require prompt medical attention.

10. References / learning sources

• World Health Organization (WHO) – Pharmacovigilance: ensuring the safe use of medicines.

• WHO–Uppsala Monitoring Centre (UMC) – The WHO-UMC system for standardised case causality assessment.

• Pharmacovigilance Programme of India (PvPI), Indian Pharmacopoeia Commission – ADR reporting and pharmacovigilance information.

Important educational disclaimer

All patient scenarios are fictional and intended only for learning. They are not real patient reports, medical diagnoses, or treatment instructions. Do not stop, restart, or change a prescribed medicine on the basis of this assignment. Clinical decisions must be made with a qualified healthcare professional.